Regime-Based Strategic Asset Allocation[†]

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Abstract

We address the question of strategic asset allocation in the presence of economic regimes. Modeling regimes as a mixture of distributions, we investigate the implications for portfolios built under popular asset allocation methodologies (mean-variance-optimization, risk budgeting). Using these analytical results, we define new portfolio construction methodologies that exploit the information in macroeconomic (macro) regimes through the composition of optimal portfolios for each regime, the risk structure of these portfolios, and the long-term probability of the regimes. Our findings have practical implications, as we empirically show that macro regime-based portfolios can outperform traditional asset-based portfolios.

Keywords: macro regimes, strategic asset allocation, mean-variance-optimization, risk-based investing.

JEL classification: G11, D81, C60.

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1 Introduction

There is ample evidence that regimes drive economies (Hamilton, 1989). The most prominent examples are recessions, generally defined as periods of decline in economic growth. Investors are concerned by recessions as they are typically associated with significant decreases in the value of their portfolios. Indeed, macroeconomic (macro) regimes explain salient characteristics of financial returns, such as changes in volatility or correlation between assets (Molenaar et al., 2024), negative skewness, or fat distribution tails (Ang and Timmermann, 2012).

How should an investor integrate the notion of regimes into asset allocation? Much literature tackles this issue by investigating how the investor can improve the returns by adapting her portfolio tactically to detected regimes. To perform these Tactical Asset Allocation (TAA) analyses, the literature has used methodologies ranging from financial returns-based regime-switching models (Ang and Bekaert, 2004; Kritzman et al., 2012) to direct macro regime classification (Ilmanen et al., 2014; Jurczenko and Teiletche, 2018; Kollár and Schmieder, 2019; de Longis and Ellis, 2023), macro business cycles (Scherer and Apel, 2020), macro nowcasting indicators (Blin et al., 2021), macro factor-mimicking portfolios (Swade et al., 2024), or machine-learning models (Mueller-Glissmann and Ferrario, 2024). Recently, Elkamhi et al. (2023) have suggested a novel approach where the TAA is implemented through investors' views on the likelihood of macro regime probabilities instead of the full spectrum of assets' return and risk, allowing for a reduction of the dimensionality curse of the parameters to be estimated.

Our article also stresses the importance of macro regimes for asset allocation but tackles the question from a strategic and analytical point of view. To many investors, strategic asset allocation (SAA) is more relevant than TAA because they do not believe in TAA's added value (Asness, 2016) or lack the resources to implement it. A regime-based approach to SAA has fundamental advantages. It notably expands the range of decisions by considering more variety in the number of economic and market scenarios (typically adverse ones). Still, it remains easy to implement and intuitive in design. We model economic regimes as a mixture of distributions. The weight attributed to each distribution is constant, assuming the investor adopts a long-term strategic policy. Using established results on the implications of mixtures of distributions for individual moments, we derive analytical expressions on the impact of regimes on asset portfolio moments and the risk contributions in two dimensions (regimes and assets). We derive implications for popular asset allocation methodologies such as mean-variance-optimization (Markowitz, 1952) or risk budgeting (Roncalli, 2013). We introduce new portfolio construction methodologies incorporating macro regime information into SAA, depending on investors' preference for capital or risk allocation and their estimates of the regimes' probabilities. We investigate the properties of these portfolio construction methodologies in in-sample and out-of-sample studies for a typical multi-asset universe over more than 50 years.

Our work is related to the few studies that have tackled the issue of regime-based investing from a SAA perspective. Clarke and de Silva (1998) show that the "state-dependent" portfolio, obtained "by systematically altering the portfolio allocation in response to changes in the investment opportunity set as the economy moves from one state to another", dominates a static (fixed parameters) portfolio. A key hypothesis is that the investor knows precisely when each regime prevails. We extend the analysis to alternative set-ups regarding investors' knowledge about regimes. Kelliher et al. (2022) empirically analyze a risk parity strategic asset allocation for which the building blocks are based on four predetermined macro portfolios and where regimes are estimated as a Hidden Markov-switching model applied to primary market factors. Our article differs by considering a broader set of strategic portfolios beyond risk parity and analytically investigating their properties. Campani et al. (2021) develop analytical solutions for optimal portfolio strategies in a continuous-time regime-switching economy with unobservable states. We obtain consistent analytical results within a simpler framework (discrete time, constant transition probabilities), and expand their research by broadening the analysis to asset allocation methodologies other than mean-variance.

The paper is organized as follows. Section 2 provides analytical results for the distribution of asset and portfolio returns in the presence of regimes. Section 3 analytically investigates the implications of regimes for asset allocation methodologies based on mean-variance optimization or risk budgeting, and devises alternative portfolio construction methodologies based on the regimes. Section 4 empirically compares these portfolio construction methodologies among themselves and versus traditional benchmarks in in-sample and out-of-sample studies. The fifth section offers a conclusion. Additional analytical results are relegated to the Appendix.

2 Return Distribution with Regimes

To model economic regimes, we assume that asset returns follow a finite mixture model. Each observation of the $K \times 1$ vector of asset excess returns $\mathbf{R}_t$, $t = 1, ..., T$, is taken from a mixture of N distributions, with respective proportions $(p_1, ..., p_N)$, where $\sum_{i=1}^{N} p_i = 1, p_i \geq 0$, and $i = 1, ..., N$. Hence, N denotes the number of regimes, while K is the number of assets. In our framework, the proportions p_i correspond to unconditional, long-term, economic regime probabilities. This type of mixture of distribution modeling has been frequently used in risk modeling (Kim and Finger, 2000; Pochon and Teiletche, 2006). The model can be seen as simplifying more sophisticated econometric models with stochastic and time-varying probabilities, such as Markov-switching models. This simplification allows us to derive tractable analytical results. In this section, we focus on the characteristics of the distribution of returns at a single asset and portfolio levels before moving to asset allocation implications in the next section.

2.1 Single-Asset Case

We begin by analyzing the first two moments for one asset, $K = 1$. In this case, the mean and variance across multiple regimes (multi-regime) are the following:

$$\mu = \sum_{i=1}^{N} p_i \mu_i, \tag{1}$$

$$\sigma^2 = \sum_{i=1}^{N} p_i \sigma_i^2 + \sum_{i=1}^{N} p_i (\mu_i - \mu)^2. \tag{2}$$

μ is the expected return in excess of cash, and σ the multi-regime volatility; μ_i and σ_i are the equivalent metrics in regime i only. The multi-regime expected return equals the probability-weighted average of the expected returns for each regime. Conversely, the variance is not simply obtained as the probability-weighted sum of the variances but also incorporates an additional term reflecting the probability-weighted average of squared deviations between the average return in each regime vs the multi-regime average. Observing variation of average returns across regimes is a usual pattern. This implies that, generally, the multi-regime variance of returns will be higher than the weighted average of the variances in the individual regimes, as the latter term in (2) is positive.

Through the ratio of (1) with (2), we obtain the multi-regime Sharpe ratio $S = \frac{\mu}{\sigma}$ as (see Appendix A):

$$S = \sqrt{\sum_{i=1}^{N} p_i \frac{\sigma_i^2}{\sigma^2}(1 + S_i^2) - 1}, \tag{3}$$

where $S_i = \frac{\mu_i}{\sigma_i}$ is the Sharpe ratio in regime i. Hence, the multi-regime Sharpe ratio is a weighted average of the Sharpe ratios in each regime, where the weights are a function of the probability of each regime and the relative risk of each regime vs the multi-regime. Sharpe ratios for regimes characterized by a higher probability of occurrence or a higher risk will have a bigger influence on the multi-regime Sharpe ratio. For the special case where regimes are equally weighted ($p_i = N^{-1}$ for all i), and variance is constant across regimes ($\sigma_i^2 = \sigma^2$ for all i), the multi-regime Sharpe ratio is equal to the quadratic mean of the Sharpe ratios in the different regimes. The expression (3) shows that the relationship between the multi-regime Sharpe ratio and the individual regimes' Sharpe ratios is complex. In particular, we cannot draw general conclusions regarding the comparison between the multi-regime Sharpe ratio, S, and the probability-weighted average Sharpe ratio across regimes, $\sum_{i=1}^{N} p_i \, S_i$ [1].

2.2 Multi-Asset Portfolios

We next analyze the multivariate case, i.e., for a portfolio of K assets ($K > 1$) whose weights are given by the $K \times 1$ vector $\boldsymbol{w}$. We use boldface to identify matrices or vectors, while scalars are left unbold; e.g., $\boldsymbol{\mu}$ vs μ. The expected return of the portfolio is given by $\mu_p = \boldsymbol{w}^T \boldsymbol{\mu}$ where the $K \times 1$ vector of expected returns is given by $\boldsymbol{\mu} = \sum_{i=1}^{N} p_i \boldsymbol{\mu}_i$ similarly to the one-asset case (1). The variance of the portfolio is given by $\sigma_p^2 = \boldsymbol{w}^T \boldsymbol{\Omega} \boldsymbol{w}$, where the covariance matrix $\boldsymbol{\Omega}$ is obtained as:

$$\boldsymbol{\Omega} = \sum_{i=1}^{N} p_i \left(\boldsymbol{\Omega}_i + (\boldsymbol{\mu}_i - \boldsymbol{\mu}) (\boldsymbol{\mu}_i - \boldsymbol{\mu})^\top \right). \tag{4}$$

The covariance matrix can be decomposed into two parts:

$$\boldsymbol{\Omega} = \boldsymbol{\Omega}_{\text{Intra}} + \boldsymbol{\Omega}_{\text{Inter}}, \tag{5}$$

$$\boldsymbol{\Omega}_{\text{Intra}} = \sum_{i=1}^{N} p_i \boldsymbol{\Omega}_i, \qquad \boldsymbol{\Omega}_{\text{Inter}} = \sum_{i=1}^{N} p_i \, (\boldsymbol{\mu}_i - \boldsymbol{\mu}) \, (\boldsymbol{\mu}_i - \boldsymbol{\mu})^\top. \tag{6}$$

[1] We discuss further in Appendix A the conditions affecting the difference between the multi-regime Sharpe ratio and the probability-weighted one.

$\boldsymbol{\Omega}_{\text{Intra}}$ is the *intra-regime* covariance, which equals the probability-weighted average of the covariances for each regime. $\boldsymbol{\Omega}_{\text{Inter}}$ is the *inter-regime* covariance, which equals the probability-weighted cross-section deviations of the expected returns for each regime versus the multi-regime expected returns. The existence of these two distinct variance-covariance matrix components is similar to what we observe in (2) for a single asset.

Specific cases lead to simpler expressions:

- if the expected returns are constant across the N regimes, we have $\boldsymbol{\mu}_i = \boldsymbol{\mu}\ \forall i$, then $\boldsymbol{\Omega}_{\text{Inter}} = \mathbf{0}$ and $\boldsymbol{\Omega} = \boldsymbol{\Omega}_{\text{Intra}}$, i.e., the multi-regime covariance equals the intra-regime covariance;

- if the covariance is constant across the N regimes, we have $\boldsymbol{\Omega}_i = \bar{\boldsymbol{\Omega}}$ for $i = 1, ..., N$, then $\boldsymbol{\Omega} = \bar{\boldsymbol{\Omega}} + \boldsymbol{\Omega}_{\text{Inter}}$, i.e., the multi-regime covariance equals the covariance for each regime plus the inter-regime covariance;

- if both the expected returns and the covariance are constant across the N regimes, we naturally obtain the standard case without regimes, $\boldsymbol{\Omega} = \bar{\boldsymbol{\Omega}}$.

Contrary to the single-asset case, we cannot draw clear conclusions on whether the multi-regime covariance coefficients are larger than the simple probability-weighted average of the individual regimes covariance coefficients, as the inter-regime term incorporates cross-asset terms that might be negative. Note, however, that the result of higher multi-regime variances (i.e., the diagonal of the covariance matrix) discussed above remains true. It is also possible to disentangle the variance and correlation coefficients by rewriting the covariance matrix as $\boldsymbol{\Omega} = \boldsymbol{D}^{1/2}\boldsymbol{C}\boldsymbol{D}^{1/2}$, with $\boldsymbol{D}$ a diagonal matrix with variances on the diagonal and $\boldsymbol{C}$ the correlation matrix. By using the same notation for each regime, $\boldsymbol{\Omega}_i = \boldsymbol{D}_i^{1/2}\boldsymbol{C}_i\boldsymbol{D}_i^{1/2}$, we get:

$$\boldsymbol{C} = \boldsymbol{D}^{-1/2}\left(\sum_{i=1}^{N} p_i \boldsymbol{D}_i^{1/2}\boldsymbol{C}_i\boldsymbol{D}_i^{1/2} + \boldsymbol{\Omega}_{\text{Inter}}\right)\boldsymbol{D}^{-1/2}. \tag{7}$$

The formulation in (7) shows that the probability-weighted correlation coefficients can be either bigger or smaller than their multi-regime counterparts. This is determined by the *inter-regime* covariance terms which ultimately depend on the variation of expected returns across regimes, as discussed earlier. A notable result is obtained when both expected returns and volatilities remain constant across regimes, in which case the multi-regime correlation equals the sum of probability-weighted correlations, i.e., $\boldsymbol{C} = \sum_{i=1}^{N} p_i \boldsymbol{C}_i$.

3 Optimal Portfolios with Regimes

In this section, we analyze the determination of optimal portfolios in the presence of economic regimes modeled as mixtures. We first investigate the relationship between the multi-regime Markowitz (1952) mean-variance-optimal (MVO) portfolio and the equivalent portfolios in the individual regimes - called "individual-regime portfolios" below. As investors increasingly use risk-based portfolios, we also investigate the implications of regimes for risk budgeting. We finally introduce new portfolio construction methodologies that combine the individual-regime portfolios in different ways.

3.1 Mean-Variance-Optimization With Regimes

We use $\boldsymbol{w}_{\text{MVO}}$ to denote the multi-regime mean-variance-optimal portfolio, and $\boldsymbol{w}_{\text{MVO},i}$ the equivalent MVO portfolio for regime i. They are the portfolios maximizing the Sharpe ratio, over the full sample (multi-regime) or specific regime i, respectively. Assuming that the investor risk-aversion γ remains constant across regimes, both portfolios are defined as:

$$\boldsymbol{w}_{\text{MVO}} = \frac{1}{\gamma}\boldsymbol{\Omega}^{-1}\boldsymbol{\mu}, \tag{8}$$

$$\boldsymbol{w}_{\text{MVO},i} = \frac{1}{\gamma}\boldsymbol{\Omega}_i^{-1}\boldsymbol{\mu}_i, \tag{9}$$

Using (1), we can rewrite (8) as:

$$\boldsymbol{w}_{\text{MVO}} = \frac{1}{\gamma}\boldsymbol{\Omega}^{-1}\sum_{i=1}^{N} p_i\boldsymbol{\mu}_i. \tag{10}$$

Comparing with (9), we obtain:

$$\boldsymbol{w}_{\text{MVO}} = \boldsymbol{\Omega}^{-1}\sum_{i=1}^{N} p_i\boldsymbol{\Omega}_i\boldsymbol{w}_{\text{MVO},i}. \tag{11}$$

The multi-regime MVO portfolio is thus a weighted average of the MVO portfolios for the different regimes. The weight given to each of these individual-regime portfolios is equal to the unconditional probability of the regime times the ratio of the covariances in the specific regime versus the full period, $\boldsymbol{\Omega}^{-1}\boldsymbol{\Omega}_i$.

To investigate these differences further, we can consider two specific cases. First, if the expected returns are constant across the N regimes, $\boldsymbol{\mu}_i = \boldsymbol{\mu}\ \forall i$, then the multi-regime MVO portfolio becomes[2]:

$$\boldsymbol{w}_{\mathrm{MVO}} = \frac{1}{\gamma}\left(\sum_{i=1}^{N} p_i \boldsymbol{\Omega}_i\right)^{-1}\left(\sum_{i=1}^{N} p_i \boldsymbol{\mu}_i\right). \tag{12}$$

In such a case, the multi-regime MVO is based on simple probability-weighted averages of individual-regime expected returns and covariances. This way of proceeding is intuitive for investors and is frequently used in the literature (Elkamhi et al., 2023; Kritzman et al., 2023). However, our previous results show that it is only valid when expected returns are constant across regimes.

The second specific case assumes that the covariance matrices are diagonal (i.e., correlation coefficients are equal to zero outside the diagonal). In this case, (11) simplifies to:

$$w_{\mathrm{MVO},(j)} = \sum_{i=1}^{N} p_i \frac{\sigma^2_{i(j)}}{\sigma^2_{(j)}}\, w_{\mathrm{MVO},i(j)}, \tag{13}$$

with $w_{\mathrm{MVO},(j)}$ and $w_{\mathrm{MVO},i(j)}$ the MVO weight for asset j, and $\sigma^2_{(j)}$ and $\sigma^2_{i(j)}$ the variance for asset j, respectively for the multi-regime and for the regime i. Expression (13) shows that the multi-regime MVO weights are equivalent to regime-probability weighted individual regime MVO weights, modulated by the fluctuation of asset variance across regimes. This result suggests alternative ways to obtain risk-based portfolios, such as the ones discussed below.

One can notice that the previous results are obtained under specific hypotheses. Firstly, we assume that the investor risk aversion remains constant over time. In Appendix B, we relax this hypothesis and show that the formula (11) is not materially modified. In Appendix B, we also derive the results under the hypothesis that the investor builds the portfolio with a volatility target, rather than based on a risk aversion hypothesis, as this might be more realistic for investors. Secondly, we can observe that the results in this section are obtained assuming that investors have no short-selling constraints. Relaxing this hypothesis does not allow us to obtain closed-form solutions. In the empirical section, we can, however, impose such constraints, allowing us to investigate how these results unfold in more constrained set-ups. Before this, we pursue the analytical analysis of regime implications, now for risk-budgeting approaches.

[2]This formulation is equivalent to the continuous-time myopic strategy obtained by Campani et al. (2021).

3.2 Risk Budgeting With Regimes

Risk budgeting has become a standard practice for institutional investors[3]. In this section, we revisit the previous results from the angle of an investor who uses risk budgeting allocation rather than capital allocation. Traditional risk budgeting is based on the decomposition of total portfolio volatility into the sum of the individual risk contributions for each asset:

$$\sigma_p = \sum_{j=1}^{K} w_j \frac{RC_j}{\sqrt{\boldsymbol{w}^\top \boldsymbol{\Omega} \boldsymbol{w}}}. \tag{14}$$

RC_j is asset j's (marginal) risk contribution, i.e., the sensitivity of the total risk to a small increase in the j-th asset portfolio weight. It is defined as $RC_j = [\boldsymbol{\Omega}\boldsymbol{w}]_j$, where $[.]_j$ denotes the j-th row of the vector in brackets. The decomposition in (14) is the foundation of risk-budgeting. For instance, the popular risk parity / equal risk contribution strategy (Maillard et al., 2010) consists of setting the portfolio weights, w_j, so that the total risk contributions are all equal across assets, $w_j RC_j = k\ \forall j$, for k a constant (and given a total portfolio weight constraint).

What happens in the presence of regimes? Replacing the covariance matrix $\boldsymbol{\Omega}$ by (4), we obtain that the risk contributions (14) can be rewritten as:

$$\sigma_p = \sum_{i=1}^{N} p_i \frac{RC_{i.}}{\sqrt{\boldsymbol{w}^\top \boldsymbol{\Omega} \boldsymbol{w}}}, \tag{15}$$

where individual regime risk contributions $RC_{i.}$ are defined as:

$$RC_{i.} = \sum_{j=1}^{K} w_j RC_{ij}, \tag{16}$$

with $RC_{ij} = \left[\left(\boldsymbol{\Omega}_i + (\boldsymbol{\mu}_i - \boldsymbol{\mu})(\boldsymbol{\mu}_i - \boldsymbol{\mu})^\top\right)\boldsymbol{w}\right]_j$. To get (15), we use the linear homogeneity of the volatility operator underlying the risk budgeting principle in (14), but, in this instance, it is also relative to regimes in addition to assets. Risk budgeting can thus be applied to other dimensions than assets. In the next subsection, we define portfolios exploiting risk budgeting principles using information in the regimes dimension[4].

[3] See Roncalli (2013) for an introduction to risk budgeting.

[4] The idea of applying risk budgeting in other dimensions than across assets has been investigated in different contexts than regimes, such as for principal component portfolios (Lohre et al., 2014) or factors (Koumou, 2023).

3.3 Incorporating Regime Information into Portfolio Construction

What about if an investor tries to incorporate the information that regimes determine asset returns, in a strategic way? From the previous theoretical results, we propose four portfolio construction methodologies based on two steps.

In the first step, we build "individual-regime" portfolios. These N portfolios, noted $\boldsymbol{w}_{\mathrm{MVO},i}$, are obtained as MVO portfolios with no-short-selling and no-leverage constraints for each regime. The second step aggregates these individual-regime portfolios, by using different methodologies. In general terms, these aggregated "regime-based" portfolios are obtained as:

$$\boldsymbol{w}_{\mathrm{rg}} = \sum_{i=1}^{N} \theta_i \boldsymbol{w}_{\mathrm{MVO},i}, \tag{17}$$

where the combination weights, θ_i, depend on the methodology adopted. One can obtain the decomposition of the volatility for these portfolios as follows:

$$\sigma_p = \sum_{i=1}^{N} \theta_i \frac{RC_i}{\sqrt{\boldsymbol{\theta}^\top \boldsymbol{\Omega}_m \boldsymbol{\theta}}}, \tag{18}$$

where $\boldsymbol{\Omega}_m$ is the $N \times N$ variance-covariance matrix of the individual-regime portfolios, $\boldsymbol{\theta}$ is the $N \times 1$ vector of weights given to the individual-regime portfolios, and $RC_i = [\boldsymbol{\Omega}_m \, \boldsymbol{\theta}]_i$ is the i-th individual-regime portfolio (marginal) risk contribution. The risk decomposition across individual-regime portfolios (18) parallels the more traditional risk decomposition across assets given in (14).

Turning to the portfolio construction methodologies, the first two are based on capital allocation:

- "rg-PWA" (for "Probability Weighted Allocation") weights the individual-regime portfolios based on their long-term probabilities, $\theta_i = p_i$. The global intuition underlying this portfolio construction is to hold a combination of assets that performs "well" on average across regimes, the average being based on the probability of each regime.

- "rg-EWA" (for "Equally Weighted Allocation") performs a simple equal-weight of the N individual-regime portfolios, $\theta_i = \frac{1}{N}$. In this case, we assume that the investor has confidence in her ability to determine individual-regime portfolios but not in assessing their probability; hence, she prefers to use a probability-agnostic aggregation. To some extent, the investor might also be motivated in replicating the empirical success of the "1/N" strategy (DeMiguel et al., 2009), but this time in a regime-based framework.

The other two regime-based portfolios use risk-budgeting principles:

- "rg-ERC" (for "Equal Risk Contribution") allocates among the individual-regime portfolios so that their risk contributions to the combined portfolio are the same. The idea is thus to follow a philosophy similar to the "All Weather" strategy but incorporating more stringent portfolio constraints. As it implements a "1/N" investment policy, it is naturally related to "rg-EWA" but differs through its definition in risk-allocation terms rather than capital-allocation terms. We use volatility as the risk measure and compute the rg-ERC weights as the $N \times 1$ vector of weights $\boldsymbol{\theta}$ minimizing the function $f_{d-ERC}(\boldsymbol{\theta}) = \sum_{i=1}^{N} \left(\theta_i \frac{RC_i}{\boldsymbol{\theta}^T \boldsymbol{\Omega}_m \boldsymbol{\theta}} - \frac{1}{N} \right)^2$.
- "rg-PRC" (for "Probability-weighted Risk Contribution") allocates among the individual-regime portfolios so that their risk contribution to the portfolio equals the long-term probabilities. It is computed by finding the $N \times 1$ vector $\boldsymbol{\theta}$ minimizing the function $f_{d-PRC}(\boldsymbol{\theta}) = \sum_{i=1}^{N} \left(\theta_i \frac{RC_i}{\boldsymbol{\theta}^T \boldsymbol{\Omega}_m \boldsymbol{\theta}} - p_i \right)^2$. By weighting the individual-regime portfolios with regime probabilities, it is naturally related to "rg-PWA", but again differs through its definition in risk-allocation terms rather than in capital-allocation terms.

Table 1 summarizes the portfolio construction methodologies that we have introduced. Using the composition of optimal portfolios for each regime ("individual-regime"), they seek to exploit the information associated with the regimes, either regarding the probability of the regimes or regarding the risk structure of these individual-regime portfolios.

Some other characteristics of these regime-based portfolio construction methodologies are worth mentioning. The rg-PWA differs from the traditional multi-regime MVO portfolio because it incorporates the information of variations of risk across regimes[5] . The rg-ERC portfolio has

[5] To illustrate this difference, one can compare (9) and (10). This shows that, for unconstrained portfolios (where long-short positions and leverage are authorized), the difference in the portfolio weights between rg-PWA and multi-regime MVO is as follows:

$$\boldsymbol{w}_{\text{MVO}} - \boldsymbol{w}_{\text{rg-PWA}} = \frac{1}{\gamma} \sum_{i=1}^{N} p_i \left(\boldsymbol{\Omega}^{-1} - \boldsymbol{\Omega}_{\boldsymbol{i}}^{-1} \right) \boldsymbol{\mu}_{\boldsymbol{i}}.$$

similarities with the well-known "All Weather" portfolio. This portfolio construction seeks to implement risk parity across macro regimes rather than the more common implementation across assets. The All Weather strategy can be summarized as follows[6]: (1) classifying the asset classes by their sensitivity to growth and inflation expectations (2) building optimal portfolios for each growth/inflation regime (four regimes in total) (3) allocating equal (25%) risk budget to each of these optimal portfolios (4) leveraging the portfolio to a target level of risk (volatility). A noticeable difference is that rg-ERC might be easier to implement for constrained investors as it does not involve long-short positions or leverage.

In the empirical application, we benchmark the regime-based portfolios devised in Table 1 with three popular asset-based portfolio construction methodologies: (i) "MVO" is the portfolio that maximizes the multi-regime Sharpe ratio (ii) "EWA" is the equal-weight allocation portfolio across all assets (DeMiguel et al., 2009) (iii) "ERC" equalizes the risk contributions of all assets across the multiple regimes (Maillard et al., 2010). While applied to the same universe of assets, the core difference between these asset-based benchmark portfolios and the regime-based is the fact that they allocate based on the assets' individual characteristics rather than based on their behavior in the regimes[7]. We now turn to an empirical evaluation of these different portfolio constructions.

[6]See "The All Weather Story", Bridgewater (January 2012), and Ang (2014) (chap. 14) for more details.

[7]In some way, this parallels the difference between asset-based and factor-based allocation highlighted in Idzorek and Kowara (2013), with factors here driven by the regimes.

4 Empirical Results

We illustrate the theoretical framework with a typical multi-asset allocation problem based on macroeconomic (macro) regimes. We first define the macro regimes and then analyze their empirical implications for asset returns. We next analyze optimal portfolios within each regime and optimal portfolios across regimes using the portfolio construction methodologies described in the previous section. We empirically compare their composition and behavior in-sample and perform out-of-sample empirical backtests.

4.1 Economic Regimes Definition

We consider growth and inflation regimes, as they are the most frequently used macro regimes for asset allocation exercises. We adopt simple and practical definitions of those regimes for the U.S. economy.

We rely on NBER recession dates for growth and qualify contractions/recessions as "low growth" and expansions as "high growth". Based on the sample used in this study (monthly observations over the 1973-2023 period), the unconditional probability for low growth and high growth are 13% and 87%, respectively.

For inflation regimes, there is no equivalent readily available definition such as NBER. Recent literature (Neville et al., 2021; Borio et al., 2023; Kritzman et al., 2023) adopt a level of 5% in year-over-year change in consumer prices as the reference threshold to define an inflation shock. Baltussen et al. (2023) adopt a 4% threshold to define high inflation, but show that their results are unchanged when using a 5% threshold. Previous research also shows the importance of decomposing the level of inflation between expected and unexpected (so-called "surprise") inflation components, as asset price reactions significantly differ across them (Neville et al., 2021). Assuming extrapolative inflation expectations, we define the inflation surprise as the difference between current inflation and its level one year before. Combining both approaches, we define a "high inflation" period as characterized by a current inflation above 5% and a positive inflation surprise (positive change in inflation over one year). All other periods are characterized as "low inflation". Based on our more than fifty-year sample, the unconditional probabilities for high and low inflation are 16% and 84%, respectively.

Following previous research (Ilmanen et al., 2014; Jurczenko and Teiletche, 2018; Elkamhi et al., 2023; Swade et al., 2024), we combine growth and inflation indicators to determine four economic regimes: "Overheating" (high growth-high inflation), "Goldilocks" (high growth-low inflation), "Stagflation" (low growth-high inflation), "Downturn" (low growth-low inflation). Figure 1 represents the historical occurrence of these macro regimes. Their long-term (unconditional) probabilities are respectively equal to 11.1%, 75.6%, 5.1%, and 8.2%. There is a large predominance of Goldilocks periods, which can also be qualified as the economy's steady state. Conversely, other regimes correspond to economic shocks happening one-fourth of the time, sometimes occurring for only a short period of time. We next analyze the implications of these regimes for individual assets and portfolio return distribution.

4.2 Asset Returns Analysis

We consider the monthly returns of seven popular asset classes for investors over the period spanning from February 1973 to December 2023: equities (EQU) as measured by the S&P 500 Index, treasuries (GOV) as measured by the Bloomberg U.S. Treasury Index, credit (CORP) as measured by the Bloomberg U.S. Corporate Investment Grade Index, inflation-linked bonds (TIPS) as measured by the Bloomberg U.S. Treasury Inflation-Protected Securities Index, listed real estate (REITs) as measured by the FTSE NAREIT All Equity REITs Index, Gold (GOLD) as measured by the Bloomberg Commodity Gold Index, and Commodities excluding Gold (COMMx) as measured by the Bloomberg Ex-Gold Commodity Index. In total, we have 611 monthly returns[8].

Table 2 reports each asset's average returns, volatility, and Sharpe ratio in the different regimes, and in the multi-regime case. We also report the weighted-average metrics where the weights are based on the unconditional probabilities of the macro regimes. Looking across economic regimes,

[8]The TIPS index only started in the first quarter of 1997 when U.S. inflation-linked bonds were issued for the first time. To backfill the returns before that date, we follow the methodology developed by Swinkels (2018) using the Survey of Professional Forecasters of the Federal Reserve Bank of Philadelphia to measure inflation expectations. The Bloomberg Ex-Gold Commodity subindex started in January 1998. Before that date, we backfilled the returns through a regression model of the global commodity index on the gold index, where the reconstructed commodities Ex-Gold index is estimated as the residual of the regression.

we observe a substantial variation in the asset return characteristics. Goldilocks (high growth-low inflation) is the best environment across assets, as this is the sole regime where all assets deliver positive returns. It is also the best for equities and REITs in terms of average returns and Sharpe ratio. Downturn is the best government and corporate bonds environment in terms of average returns, but only for government bonds in terms of Sharpe ratio. Gold and commodities ex-Gold both display their highest performance during high inflation regimes. However, the highest returns are achieved when high inflation is combined with low growth (stagflation) for Gold, while this is with high growth (overheating) for other commodities. Both types of commodities also differ materially in downturn. Inflation also triggers different behaviors of traditional equities vs REITs, with the latter outperforming during high inflation regimes. Finally, we observe that volatilities vary primarily according to the growth regime, as for all assets, they tend to be substantially higher (with frequently more than 50% increase) in stagflation and downturn.

Table 3 displays all the correlation pairs in the different regimes and over the full sample. Here again, we observe some disparities across regimes. On average, across assets, we observe that correlation coefficients are higher during downturn periods. As measured by the standard deviation across regimes, the correlation coefficients that fluctuate the most involve inflation-exposed assets such as TIPS, COMMx, or REITS, as stagflation shocks notably trigger changes in correlation signs. Conversely, the most stable correlation coefficients are observed within fixed income (GOV, TIPS, CRED).

Based on our analytical results in Section 2, a natural question is how the statistics aggregate across the regimes. A simple way to look at this is to compare the multi-regime statistics with a regime probability-weighted average. Looking at the relevant rows and columns in Table 2 and Table 3, we have the following conclusions. As one can expect from Equation (1), average returns aggregate simply across regimes so that average returns across all regimes are equal to the probability-weighted average of average returns in individual regimes.

We also know from the analytical results that this result shall not hold for volatility (see Equation (2)). This is indeed what we observe in the empirical results: Probability-weighted averages of volatilities are consistently lower than their long-term values, with absolute differences ranging from 0.1% for government bonds to 1% for REITs. This empirical pattern is expected from the analytical results as long as average returns vary across regimes. For Sharpe ratios, the relationship between the multi-regime Sharpe ratio and the probability-weighted one is complex (see Equation (3)). Empirically, we observe that the multi-regime Sharpe ratio is systematically lower than the probability-weighted average, except for Gold. A noticeable difference is that contrary to other assets, Gold displays a lower Sharpe ratio in the most probable and lowest volatility regime (goldilocks). As we discuss in Appendix A, this is a critical condition for the hierarchy of multi-regime and probability-weighted (squared) Sharpe ratios.

For correlation coefficients, we also know from the analytical results that there is no clear hierarchy between the multi-regime metrics and their probability-weighted averages. Indeed, this is what we observe empirically (Table 3): For some pairs of assets, multi-regime correlation coefficients are higher, while in some other cases, their probability-weighted counterparts are higher. We will next investigate how these changes in asset metrics across economic regimes affect portfolio construction and asset allocation.

4.3 (Macro) Individual-Regime Portfolios Composition

As explained before, the first step of our methodology consists of building "individual-regime" portfolios, i.e., optimal portfolios for each macro regime. For this, we compute long-only mean-variance-optimal (MVO) portfolios based on the average returns and the covariance matrix in each regime[9].

Table 4 shows the composition of the individual-regime portfolios. The asset class weights vary substantially across all the macro regimes. No asset is represented in every individual-regime portfolio, but all assets are part of at least one individual-regime portfolio. Gold, commodities ex-gold, and REITs are the sole assets represented in two distinct macro regimes, notably due to their better returns during inflationary periods (Table 2). Despite their high correlation (Table 3), fixed-income assets are each associated with distinctive individual regimes: downturn for nominal government bonds, goldilocks for corporate bonds, and stagflation for inflation-linked bonds. The

[9] An alternative approach to design macro individual-regime portfolios could be based on the growing literature on macro factor-mimicking portfolios (Chousakos and Giamouridis, 2020; Swade et al., 2021; Esakia and Goltz, 2023; Jurczenko and Teiletche, 2023). See Swade et al. (2024) for an application to TAA.

most diversified individual-regime portfolio, in terms of number of assets incorporated, is the Goldilocks portfolio. The Overheating portfolio is composed of three assets: Gold, commodities ex-gold, and REITs, while the Stagflation portfolio involves two assets, TIPS and Gold. The Downturn portfolio is the sole portfolio concentrated on one asset only, with government nominal bonds.

In Table 4, we also report the macro-regime portfolios' risk-return metrics in their individual macro regime and for the full sample (i.e., across all macro regimes). The macro-regime portfolios all display much higher Sharpe ratios in their individual regime than the multi-regime Sharpe ratio. This intuitive result indicates that these portfolios may not be robust across the various regimes. In addition, the concentration of some of these portfolios leads to high volatility, so they seem unrealistic for many investors to hold strategically. In this perspective, we next investigate portfolios that consider macro regime information but are diversified enough to be held strategically.

4.4 (Macro) Regime-Based Portfolios Backtesting

We investigate four regime-based portfolios aggregating the macro individual-regime portfolios given in Table 4: (i) "rg-PWA" weights the individual-regime portfolios through their long-term probabilities; (ii) "rg-EWA" equally weights the four individual-regime portfolios; (iii) "rg-ERC" equalizes the risk contributions of the individual-regime portfolios; (iv) "rg-PRC" aligns the individual-regime portfolios risk contributions with their long-term probabilities. To benchmark these regime-based portfolios, we use three asset-based portfolios: the multi-regime "MVO" portfolio, the equal-weight "EWA" portfolio, and the equal risk contribution "ERC" portfolio.

Figure 2 shows the asset allocation corresponding to these seven (four regime-based and three asset-based) portfolios. We observe significant differences in portfolio weights across the methodologies. In particular, rg-PWA and MVO, rg-EWA and EWA, or rg-ERC and ERC are quite different despite their common portfolio construction features. Although individual-regime portfolios tend to be concentrated in a few assets (Table 3), the weighting mechanisms across regimes lead to portfolios invested in *all* assets. The associated diversified profiles notably differ from the multi-regime MVO portfolio, which is not invested in some assets (for instance, corporate bonds and inflation-linked bonds are excluded) and is highly concentrated in government nominal bonds (more than 50% weight).

In Table 5, we investigate the implications of these different portfolio construction methodologies for performance and risk, both for the individual macro regimes and the full sample (multi-regime). The portfolio with the highest multi-regime Sharpe ratio is, by construction, the MVO. Among the asset-based portfolios, it is also the most performing, while ERC is more defensive. Turning to macro regime-based portfolios, we see that the best-performing multi-regime portfolio is rg-PWA, which weighs the individual-regime portfolios based on long-term probabilities. As discussed before, this portfolio is close to MVO: they are similar when the risk does not change across regimes. Looking at the tracking error volatility (TEV) vs the MVO portfolio, rg-PRC is, however, closer to MVO than rg-PWA, as can also be observed in Figure 2. This means that, for this specific sample, incorporating long-term macro-regime probabilities as risk budgets rather than capital weights aligns better with the traditional MVO portfolio. Looking at the behavior in the different regimes, we observe that the equal risk contribution portfolio across regimes, rg-ERC, posts a regular behavior in performance across the different regimes. Indeed, it posts the highest risk-adjusted returns in downturn and stagflation and, respectively the highest and the second highest absolute returns in these regimes, while it is fourth in overheating. It is thus a robust portfolio across the various regimes outside goldilocks.

In Table 6, we report additional multi-regime risk metrics, i.e., maximum drawdown, skewness, worst month, and stress tests computed as the performance during specific financial crisis (October 1987, August 1998, October 2008, March 2020). These four months are the worst months observed for a 60/40 portfolio over our sample. They correspond to well-known shocks: Black Monday, the Asian crisis, the Global Financial Crisis, and COVID-19. The robustness of the rg-ERC portfolio during bad times is visible across all metrics. Indeed, it is the portfolio displaying the lowest maximum drawdown, the least negative skewness, and the least negative worst month, and it outperforms the other portfolios across all the stress test episodes. We notably observe that the regime-based equal risk portfolio, rg-ERC, is more defensive than its asset-based equivalent, ERC.

To evaluate the performance of the various portfolio constructions more generally in different set-ups, we also perform an out-of-sample test. It consists of re-estimating the portfolio compositions monthly, based on an expanding window. Every month, we re-estimate the different parameters (average returns, variances, and probabilities for the various regimes) and infer the various portfolio construction compositions based on a new sample from the first month in the sample (February 1973) to the current month. The new portfolio is next held during the following month, and the associated return is recorded. The test is out-of-sample as no information from the future is used in the estimation. We apply this methodology from the half to the end of the sample (June 1998 to December 2023)[10].

We summarize the results of this out-of-sample test in Table 7. The dominance of rg-ERC is confirmed. From a risk perspective, it remains the most robust portfolio construction, posting the lowest maximum drawdown, the least negative skewness, or the lowest worst month. Interestingly enough, we observe that these risk robustness properties also have significant implications on out-of-sample risk-adjusted returns. In particular, rg-ERC is the portfolio with the highest Sharpe ratio, notably surpassing *out-of-sample* MVO which is by construction the portfolio construction with the highest *in-sample* Sharpe ratio.

To further understand the risk-return trade-off of the different portfolios out-of-sample, we follow Boudoukh et al. (2019) and compute the Certainty Equivalent Return (CER) of portfolio p against the MVO benchmark as $CER_p = \mu_p - \mu_{\text{ MVO}} - 0.5\,\gamma\left(\sigma_p^2 - \sigma^2_{\text{MVO}}\right)$. The CER expresses the additional risk-adjusted return obtained by a quadratic utility function investor when she selects any of the portfolios other than MVO. As in Boudoukh et al. (2019), we use both $\gamma = 3$ (low risk-aversion) and $\gamma = 10$ (high risk-aversion). As shown in Table 7, rg-ERC has the highest CER metrics for risk-averse investors ($\gamma = 10$), meaning such type of investors would always prefer rg-ERC over all other portfolio construction, whether regime-based or asset-based.

[10] We chose half of the sample as the starting point of the out-of-sample period to ensure we have enough observations to estimate the parameters, and in particular enough observations of the various regimes.

5 Conclusion

What should investors do in the presence of economic regimes? Researchers and practitioners usually address this topic from a tactical asset allocation point of view. In this article, we depart from the literature by tackling the issue strategically, which might be more relevant for many investors.

Modeling economic regimes as a mixture of distributions, we have determined the implications for portfolios built under popular asset allocation methodologies (mean-variance-optimization, risk budgeting). Based on these analytical results, we have presented new portfolio construction methodologies seeking to exploit the information in the regimes, using the composition of optimal portfolios for each regime, the risk structure of these portfolios, and, in some cases, the probability of the regimes. We have empirically shown that, for economic regimes, regime-based portfolio construction methodologies outperform assets-based ones for a typical multi-asset universe. In particular, applying an equal risk contribution principle across macro individual-regime portfolios yields a robust portfolio across macro regimes.

These promising results might be further investigated in alternative practical applications. As regimes can lead to skewness and kurtosis in the distribution of returns, it would be interesting to apply methodologies that incorporate higher moments in asset allocation, such as Jurczenko and Teiletche (2019). Other portfolio construction methodologies could also be investigated, such as minimum-regret approaches (Chan et al., 2021). We leave these developments to future research.

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Appendix. Additional Analytical Results

Appendix A. Regime Implications for Sharpe Ratios

We analyze the implications of regimes for the Sharpe ratio. We start by rewriting the variance (2) as follows:

$$\sigma^2 = \sum_{i=1}^{N} p_i(\sigma_i^2 + (\mu_i - \mu)^2) = \sum_{i=1}^{N} p_i(\mu_i^2 + \sigma_i^2 - \mu^2), \tag{A1}$$

We can infer the relationship between the Sharpe ratio of the overall period and the Sharpe ratio for each regime, as follows:

$$\begin{aligned}
\sigma^2 + \mu^2 &= \sum_{i=1}^{N} p_i \left(\sigma_i^2 + \mu_i^2\right) \\
\sigma^2 \left(1 + \frac{\mu^2}{\sigma^2}\right) &= \sum_{i=1}^{N} p_i \sigma_i^2 \left(1 + \frac{\mu_i^2}{\sigma_i^2}\right) \\
1 + \left(\frac{\mu}{\sigma}\right)^2 &= \sum_{i=1}^{N} p_i \frac{\sigma_i^2}{\sigma^2} \left(1 + \left(\frac{\mu_i}{\sigma_i}\right)^2\right).
\end{aligned} \tag{A2}$$

Using the notations $S_i = \frac{\mu_i}{\sigma_i}$ and $S = \frac{\mu}{\sigma}$ for the Sharpe ratio for regime i and for the full sample, respectively, and rearranging (A2), yields:

$$S = \sqrt{\sum_{i=1}^{N} p_i \frac{\sigma_i^2}{\sigma^2}(1 + S_i^2) - 1}. \tag{A3}$$

The expression (A3) shows that the relationship between the multi-regime Sharpe ratio and the individual regimes' Sharpe ratios is complex, and depends on the specific ratios $\frac{\sigma_i^2}{\sigma^2}$ and the distribution of the regime probabilities p_i. In particular, we cannot draw general conclusions regarding the comparison between the multi-regime Sharpe ratio, S, and the probability-weighted average Sharpe ratio across regimes, $\sum_{i=1}^{N} p_i\, S_i$.

We can analyze a related question by comparing the squared multi-regime Sharpe ratio, S^2, and the probability-weighted average squared Sharpe ratio across regimes, $\sum_{i=1}^{N} p_i\, S_i^2$. Squaring both sides of (A3) and subtracting $\sum_{i=1}^{N} p_i\, S_i^2$ yields the difference:

$$S^2 - \sum_{i=1}^{N} p_i\, S_i^2 = \sum_{i=1}^{N} p_i\, (1 + S_i^2) \left(\frac{\sigma_i^2}{\sigma^2} - 1\right). \tag{A5}$$

The sign of the difference depends on whether σ_i^2 is lower or greater than σ^2 in the different regimes. While we cannot get to general conclusions, we might expect from (A5) that the squared multi-regime Sharpe ratios will be lower than the probability-weighted average of individual squared Sharpe ratios when less volatile regimes (σ_i is low versus σ^2) are more probable (p_i is high) and have higher Sharpe ratios (S_i is high), and vice versa.

Appendix B. Additional Results on MVO Aggregation Across Regimes

In the body part of the text, we investigate the case of an investor with a constant risk aversion across regimes. As an additional development, we here first relax this hypothesis. Assuming that the risk aversion is different across regimes allows us to generalize Equation (11) as follows:

$$\boldsymbol{w}_{\mathrm{MVO}} = \frac{1}{\gamma}\boldsymbol{\Omega}^{-1}\sum_{i=1}^{N} p_i \gamma_i \boldsymbol{\Omega}_i \boldsymbol{w}_{\mathrm{MVO},i}. \tag{B1}$$

The expression (B1) is only moderately more complex than the constant risk aversion case (11). However, from a practical standpoint, it requires a different risk aversion for each regime and is thus more demanding regarding the number of inputs to estimate.

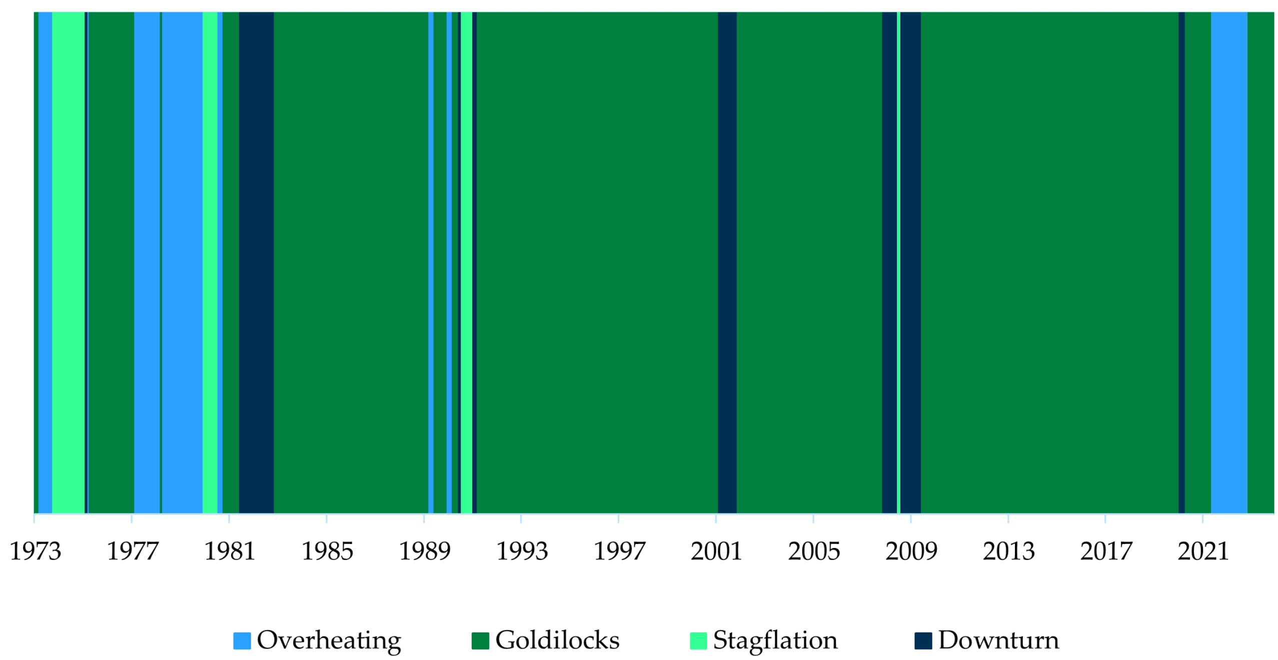
Determining an appropriate risk aversion level is a common issue when practically implementing MVO. To cope with this issue, a standard solution is to target a constant portfolio volatility, σ_T. In this case, the Sharpe ratios of MVO portfolios for the full sample and the individual regimes becomes $S = \sqrt{\mu^\top \Sigma^{-1} \mu}$ and $S_i = \sqrt{\mu_i^\top \Sigma^{-1} \mu_i}$, respectively. From this, we deduce that:

$$\boldsymbol{w}_{\mathrm{MVO}} = \sum_{i=1}^{N} p_i \frac{S_i}{S} \boldsymbol{\Omega}^{-1} \boldsymbol{\Omega}_i \boldsymbol{w}_{\mathrm{MVO},i}. \tag{B2}$$

We notice that the portfolio weights are independent of the level of the target volatility σ_T.

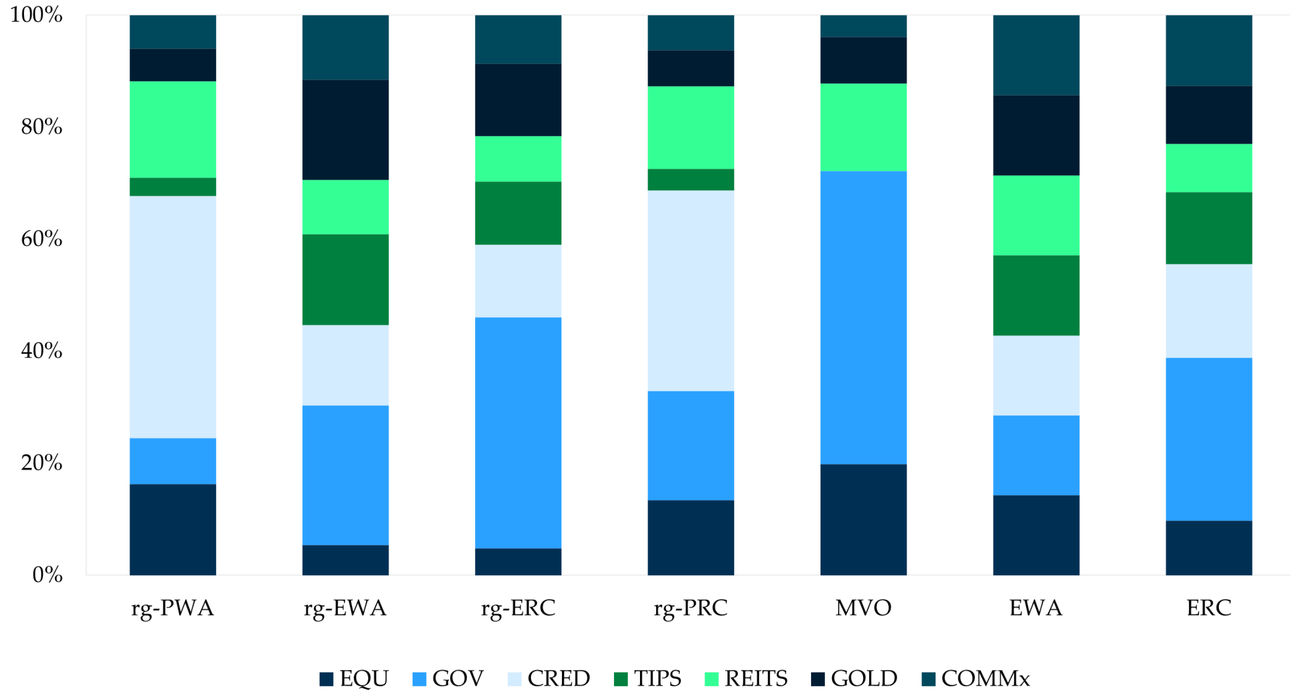
Figures & Tables

Figure 1: Economic Regimes (Growth and Inflation combined)



Notes: Overheating is characterized by high growth and high inflation, Goldilocks by high growth and low inflation, Stagflation by low growth and high inflation, and Downturn by low growth and low inflation. High inflation corresponds to current inflation above 5% and positive inflation surprises. Low growth corresponds to NBER recessions. *Sources*: NBER, Bloomberg, authors' calculations.

Figure 2: Regime-Based and Asset-Based Multi-Asset Portfolio Allocations



Notes: The figure shows the asset allocation of the various portfolios. rg-PWA, rg-EWA, rg-ERC, and rg-PRC are the four regime-based portfolios. rg-PWA weights the four individual macro-regime portfolios based on the long-term macro-regime probability. rg-EWA weighs the four individual macro-regime portfolios equally. rg-ERC equalizes the risk contribution of the four individual macro-regime portfolios. rg-PRC aligns the four individual macro-based portfolio risk budgets with the long-term macro-regime probabilities. MVO, EWA, and ERC are traditional asset-based portfolio construction methodologies based on mean-variance optimization, equal weighting, and equal risk contribution.

Table 1: Description of Regime-Based Portfolios

Portfolio	Allocation Mechanism	Individual-Regime Portfolios Weighting
rg-PWA	Capital-based	Regime-probability (p_i)
rg-EWA	Capital-based	Equal-weight ($1/N$)
rg-ERC	Risk-based	Equal-weight ($1/N$)
rg-PRC	Risk-based	Regime-probability (p_i)

Notes: The table describes the main features underlying the composition of the regime-based portfolios. They all use the "individual-regime" portfolios as building blocks, i.e., the constrained MVO portfolios obtained for each regime.

Table 2: Asset Returns Characteristics in the Different Regimes

	EQU	GOV	CRED	TIPS	REITs	GOLD	COMMx
Overheating (probability: 11.1%)							
Average return	6.1%	0.8%	-1.7%	4.6%	12.2%	26.5%	24.2%
Volatility	14.7%	5.3%	7.8%	11.6%	17.0%	26.5%	22.2%
Sharpe ratio	-0.02	-1.06	-1.03	-0.16	0.34	0.76	0.80
Goldilocks (probability: 75.6%)							
Average return	14.9%	6.2%	8.0%	6.3%	15.0%	4.0%	6.2%
Volatility	13.8%	4.6%	5.5%	8.4%	13.8%	16.1%	13.3%
Sharpe ratio	0.78	0.46	0.72	0.26	0.79	-0.01	0.16
Stagflation (probability: 5.1%)							
Average return	-0.6%	9.8%	5.0%	17.1%	0.6%	29.3%	5.1%
Volatility	23.2%	7.8%	13.4%	16.3%	23.1%	36.1%	32.1%
Sharpe ratio	-0.46	-0.03	-0.38	0.43	-0.41	0.53	-0.15
Downturn (probability: 8.2%)							
Average return	-6.2%	12.3%	11.7%	3.7%	-6.8%	10.8%	-10.8%
Volatility	22.8%	7.5%	12.5%	16.7%	34.0%	22.0%	22.1%
Sharpe ratio	-0.57	0.74	0.40	-0.18	-0.40	0.18	-0.79
Full sample (multi-regime)							
Average return	11.4%	6.3%	7.1%	6.4%	12.1%	8.3%	6.8%
Volatility	15.5%	5.2%	7.2%	10.2%	17.3%	19.5%	16.8%
Sharpe ratio	0.42	0.27	0.31	0.15	0.42	0.18	0.11
Probability-weighted							
Average return	11.4%	6.3%	7.1%	6.4%	12.1%	8.3%	6.8%
Volatility	15.1%	5.1%	6.7%	9.8%	16.3%	18.7%	16.0%
Sharpe ratio	0.52	0.29	0.44	0.18	0.58	0.12	0.14

Notes: EQU, GOV, CRED, TIPS, REITs, GOLD, and COMMx stand for equities, treasuries, credit, inflation-linked bonds, listed real estate, Gold, and commodities ex-Gold. See Figure 1 for the definition and representation of the regimes. "Probability-weighted" is the weighted average of the average returns, volatilities, and Sharpe ratios in the different regimes, where the weights are based on the macro regimes' unconditional probabilities. The sample period is February 1973-December 2023.

Table 3: Correlation between Asset Returns in the Different Regimes

Assets pair	Overheating	Goldilocks	Stagflation	Downturn	Multi-Reg.	Prob-Wgt
EQU, GOV	0.44	0.04	0.15	0.08	0.09	0.09
EQU, CRED	0.52	0.27	0.44	0.57	0.38	0.33
EQU, TIPS	0.37	0.20	-0.07	0.39	0.22	0.22
EQU, REITs	0.83	0.51	0.80	0.74	0.62	0.58
EQU, GOLD	-0.02	-0.04	0.19	0.15	0.01	-0.01
EQU, COMMx	0.16	0.19	-0.31	0.48	0.16	0.19
GOV, CRED	0.92	0.87	0.85	0.60	0.81	0.85
GOV, TIPS	0.83	0.80	0.84	0.73	0.78	0.80
GOV, REITs	0.40	0.13	0.04	0.01	0.11	0.14
GOV, GOLD	0.07	0.08	-0.06	0.25	0.07	0.08
GOV, COMMx	-0.08	-0.15	-0.07	-0.18	-0.15	-0.14
CRED, TIPs	0.77	0.79	0.71	0.80	0.77	0.78
CRED, REITs	0.51	0.33	0.23	0.46	0.36	0.36
CRED, GOLD	-0.07	0.12	-0.06	0.27	0.07	0.10
CRED, COMMx	-0.04	-0.04	-0.11	0.25	-0.02	-0.02
TIPS, REITs	0.38	0.17	-0.20	0.27	0.18	0.18
TIPS, GOLD	0.04	0.07	-0.06	0.31	0.08	0.08
TIPS, COMMx	-0.02	-0.03	0.18	0.12	0.03	0.00
REITs, GOLD	0.03	0.08	0.26	0.06	0.08	0.09
REITs, COMMx	0.14	0.17	-0.27	0.41	0.16	0.16
GOLD, COMMx	0.11	0.19	-0.05	0.12	0.13	0.17
Median	**0.16**	**0.17**	**0.04**	**0.27**	**0.13**	**0.16**

Notes: The table represents the correlation coefficient between the pair of assets listed in the first column, in the different macro regimes. See Figure 1 for the definition and representation of the regimes. "Multi-Reg." are the multi-regime (full sample) correlations. "Prob-Wgt" is the weighted average of correlation coefficients in the different regimes, where the weights are based on the macro regimes' unconditional probabilities. The sample period is February 1973-December 2023.

Table 4: Composition & Risk-Return Characteristics of Macro Individual-Regime Portfolios

	Overheating	Goldilocks	Stagflation	Downturn
Portfolio composition				
EQU	0.0%	21.5%	0.0%	0.0%
GOV	0.0%	0.0%	0.0%	100.0%
CRED	0.0%	57.2%	0.0%	0.0%
TIPS	0.0%	0.0%	64.7%	0.0%
REITs	19.1%	19.8%	0.0%	0.0%
GOLD	36.4%	0.0%	35.3%	0.0%
COMMx	44.5%	1.4%	0.0%	0.0%
Risk-return metrics in own individual regime				
Average return	22.7%	10.9%	21.4%	12.3%
Volatility	15.3%	6.8%	16.0%	7.5%
Sharpe ratio	1.07	1.00	0.71	0.74
Multi-regime risk-return metrics				
Average return	8.4%	9.0%	7.1%	6.3%
Volatility	12.0%	8.7%	9.9%	5.2%
Sharpe ratio	0.29	0.48	0.22	0.27

Notes: The table represents the composition of the mean-variance-optimized portfolios for each macro regime (so-called "individual-regime" portfolios), and their regime-specific and multi-regime risk-return metrics. The sample is February 1973-December 2023.

Table 5: Risk-Return Metrics of the Different Multi-Asset Portfolios (in-sample)

	Regime-Based				Asset-Based		
	rg-PWA	**rg-EWA**	**rg-ERC**	**rg-PRC**	**MVO**	**EWA**	**ERC**
Multi-regime							
Average return	8.6%	7.7%	7.4%	8.3%	8.4%	8.4%	7.7%
Volatility	7.6%	6.7%	5.9%	6.9%	6.6%	7.4%	6.3%
Sharpe ratio	0.49	0.42	0.44	0.50	0.54	0.47	0.46
TEV vs MVO	2.0%	3.5%	3.2%	1.6%	0.0%	2.7%	2.6%
Average Returns in Different Regimes							
Overheating	5.6%	9.7%	7.5%	5.6%	6.7%	10.4%	8.0%
Goldilocks	9.8%	7.4%	7.3%	9.2%	9.1%	8.6%	7.9%
Stagflation	5.6%	11.8%	10.9%	6.6%	7.7%	9.5%	9.6%
Downturn	4.0%	5.1%	6.6%	4.9%	4.6%	2.1%	4.6%
Risk-Adjusted Average Returns in Different Regimes							
Overheating	0.67	1.21	1.07	0.72	0.89	1.20	1.09
Goldilocks	1.63	1.34	1.49	1.67	1.67	1.44	1.54
Stagflation	0.49	1.17	1.24	0.63	0.79	0.90	1.05
Downturn	0.29	0.46	0.70	0.39	0.42	0.16	0.43

Notes: The table provides return-risk metrics for the various portfolios. rg-PWA, rg-EWA, rg-ERC, and rg-PRC are the four macro regime-based portfolios. MVO, EWA, and ERC are standard asset-based portfolio constructions using mean-variance optimization, equal weighting, and equal risk contribution, respectively. See Figure 2 for their composition. TEV is the tracking error volatility, computed as the annualized standard deviation of excess returns over MVO returns. The sample period is February 1973-December 2023.

Table 6: Additional Risk Metrics for the Different Multi-Asset Portfolios (in-sample)

	Regime-Based				Asset-Based		
	rg-PWA	**rg-EWA**	**rg-ERC**	**rg-PRC**	**MVO**	**EWA**	**ERC**
				Risk Metrics			
Maximum drawdown	26.6%	17.6%	13.5%	22.7%	20.8%	26.6%	19.1%
Skewness	-0.60	-0.50	-0.34	-0.54	-0.52	-0.84	-0.58
Worst month	-13.6%	-12.1%	-9.5%	-12.1%	-10.7%	-14.8%	-11.2%
				Stress Tests			
October 1987	-4.0%	0.3%	0.9%	-2.7%	-4.3%	-2.9%	-0.6%
August 1998	-4.1%	-2.3%	-1.3%	-3.3%	-3.5%	-4.4%	-2.5%
October 2008	-13.6%	-12.1%	-9.5%	-12.1%	-10.7%	-14.8%	-11.2%
March 2020	-8.9%	-4.5%	-3.1%	-7.3%	-4.3%	-7.2%	-5.1%

Notes: The table provides additional risk metrics for the various portfolios. See the notes in Table 5 for their description and Figure 2 for their composition. The sample period is February 1973-December 2023.

Table 7: Risk-Return Metrics for the Different Multi-Asset Portfolios (out-of-sample)

	Regime-Based				Asset-Based		
	rg-PWA	**rg-EWA**	**rg-ERC**	**rg-PRC**	**MVO**	**EWA**	**ERC**
Average return	6.0%	6.0%	5.6%	5.8%	5.3%	6.4%	5.6%
Volatility	8.0%	7.1%	6.3%	7.4%	7.3%	7.9%	6.6%
Sharpe ratio	0.51	0.57	0.58	0.52	0.46	0.56	0.56
Maximum drawdown	27.9%	19.8%	16.7%	25.4%	26.9%	26.6%	21.7%
Skewness	-1.14	-1.11	-1.01	-1.14	-1.52	-1.31	-1.32
Worst month	-13.8%	-12.8%	-10.7%	-12.7%	-14.4%	-14.8%	-12.5%
CER vs MVO ($\gamma = 3$)	0.56%	0.77%	0.54%	0.48%	0.00%	0.93%	0.50%
CER vs MVO ($\gamma = 10$)	0.21%	0.85%	1.00%	0.44%	0.00%	0.60%	0.83%

Notes: The table provides return-risk metrics for the various portfolios for the out-of-sample period (June 1998-December 2023). CER is the Certainty Equivalent Return as computed in Boudoukh et al. (2019), with γ the risk aversion coefficient.